

Replication Report: Muis et al. (2022)

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Research Design: Causal and Descriptive Analyses Using R (WS 25/26)

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Replication package: <https://github.com/santiago2108/far-right-government-replication>

The author has no conflict of interest with the original authors.

March 1, 2026

Abstract

Muis et al. (2022) examine whether far-right government inclusion moderates the protest-voting mechanism across 22 European countries using European Social Survey data from 2002–2016. Their central claim is that political distrust drives far-right voting more strongly when these parties are in opposition than when they hold office. I reconstruct the authors' core multilevel logistic models (Table 2, Models 1–3) in R using ESS Rounds 1–8, ParlGov cabinet data, and The PopuList 3.0. No original analysis code or dataset was provided; the authors shared SPSS syntax for data preparation, and neither computational reproducibility from analytic data (CRA) nor from raw data (CRR) can be assessed for the original workflow in the sense of Berkeley Initiative for Transparency in the Social Sciences (2020). I therefore perform an independent recreate reproduction, translating the SPSS syntax into R. All key coefficients reproduce in sign and magnitude. The trust $\times$ far-right-in-power interaction (OR = 1.398, 95% CI [1.329, 1.470], $p < 0.001$) confirms the inclusion–moderation hypothesis.

Keywords: replication; computational reproducibility; far right; protest voting; political trust; ESS

1 Introduction

1.1 Original study: research question and main claim

Why do citizens vote for far-right populist parties? One possible answer is the *protest-voting* hypothesis: people who distrust mainstream political institutions turn to far-right parties as a way to signal discontent with the establishment. Yet this explanation has the implicit assumption that far-right parties remain outside the corridors of power. Muis et al. (2022) ask what happens to this mechanism once far-right parties actually enter government – an increasingly common scenario in contemporary Europe and the world.

Their theoretical argument is straightforward. When a far-right party is in opposition, voting for it is a credible act of anti-elite protest. When the same party joins a governing coalition, it becomes part of the establishment it once attacked. Under these conditions, the link between political distrust and far-right voting should weaken: the party can no longer plausibly serve as a vehicle of anti-establishment sentiment. The authors label this the *inclusion–moderation* hypothesis.

The paper therefore pursues two research questions. The primary question is: to what extent does the characterization of far-right populist voters as being disillusioned with the political establishment depend on whether their parties are in power or not (Muis et al., 2022, p. 751) The secondary question extends this to a new geographical context: are previous findings supporting the inclusion–moderation thesis – based on Western Europe – equally valid in Central and Eastern European countries (CEECs) (Muis et al., 2022, p. 752)

Using European Social Survey (ESS) data from Rounds 1–8 (2002–2016) across 22 countries – spanning both Western Europe and post-communist CEECs – Muis et al. test these questions with multilevel binary logistic regression. The dependent variable is a binary indicator of whether a respondent reported voting for a far-right party in the most recent national general election, with all other voters (including blank votes) as the reference category. The key independent variable is a context-level dummy coded 1 when a far-right party participates in the national government coalition or formally supports a minority cabinet (sourced from ParlGov). The individual-level moderator is a political trust scale averaging self-reported trust in parliament, politicians, and political parties (0–10 scale). Because ESS fieldwork can span government changes within a single round, each respondent is assigned to the appropriate *country-period* based on their interview date.

Following the structured claim-summary convention recommended by Berkeley Initiative for Transparency in the Social Sciences (2020), the paper tested whether political trust (X) is associated with far-right voting (Y) among voting-age adults across 22 European countries (P),

using multilevel binary logistic regression (M).¹ The main results show a positive interaction effect of far-right government participation on the trust–vote association, with an interaction odds ratio of approximately 1.47 ($p < 0.05$), indicating that the negative association between political trust and far-right voting is attenuated when far-right parties hold government office. No 95% confidence interval for the interaction term is reported in the published table, limiting independent verification of uncertainty beyond the significance markings. Summarising the results across both Western and Central-Eastern Europe, the authors state on page 765: “political distrust is a less important determinant of electoral support when far-right parties are in power – the positive OR of the interaction-term means that the negative association between trust and far-right voting is attenuated when far-right parties are in government” (Muis et al., 2022, p. 765).

A secondary finding is that the attenuation pattern also appears in post-communist Europe: in Hungary and Poland, far-right voters are even *less* distrustful of national political institutions than the rest of the electorate when far-right parties are in power.

1.2 What this replication report does

This report assesses whether the central empirical claim of Muis et al. (2022) is computationally reproducible. Specifically, I reconstruct Models 1–3 of their Table 2 – the three-step multilevel logistic regression sequence that establishes (a) the baseline trust effect, (b) the effect net of policy-related attitudes, and (c) the interaction between far-right government inclusion and political trust – using only publicly available data and the variable-construction logic documented in SPSS syntax shared by the author (J. Muis).

All analysis is conducted in R, and no original executable analysis code was available. Because the authors did not provide an analytic dataset or executable analysis code, computational reproducibility from analytic data (CRA)² cannot be assessed for the original workflow. Likewise, the authors’ computational reproducibility from raw data (CRR) as implemented by them cannot be evaluated because no runnable end-to-end pipeline was provided. I therefore conduct an independent recreate reproduction starting from raw data: I reconstruct the data-preparation steps from the author-provided SPSS syntax and implement the full preparation and analysis pipeline in R. All R scripts, data files used in the replication, and output files are available at <https://github.com/santiago2108/far-right-government-replication>.

No extension to the original analysis is included in this report. The replication focuses entirely

¹The ACRE notation uses four components to summarise a causal or descriptive claim: X = key independent variable (here: political trust), Y = outcome variable (far-right vote), P = population or sample (voting-age adults in 22 European countries, ESS Rounds 1–8), and M = method of analysis (multilevel binary logistic regression).

²CRA (*computationally reproducible from analytic data*): the ability to reproduce the published outputs starting from the processed dataset and analysis code provided by the authors. CRR (*computationally reproducible from raw data*): the ability to reproduce the published outputs starting from the raw, unprocessed data using the authors’ own end-to-end pipeline. Neither can be assessed here because neither an analytic dataset nor executable analysis code was provided.

on whether the sign, approximate magnitude, and statistical significance of the key coefficients – above all the interaction term $\text{Trust} \times \text{Far right in power}$ – can be recovered through independent reconstruction. An extension was considered, but not pursued: the multilevel random-intercept structure required by the original specification is technically demanding to implement correctly in R at the current stage of my training. Although a country fixed-effects specification would have been more tractable, I opted to retain the original multilevel structure to maximise comparability with the published results. The fixed-effects model is instead included as a sensitivity check in Section 3.3.

1.3 Data sources and transparency

The individual-level data come from the European Social Survey, Rounds 1–8, downloaded directly from the ESS Data Portal. The context-level government participation variable is constructed from the ParlGov database, cross-validated against The PopuList 3.0 for party classification. Cabinet assignment follows the author-provided SPSS logic, matching each respondent’s interview date to the cabinet in office at that time. All data preparation and analysis scripts are written in R and are available together with this report. Full variable coding rules are reported in Appendix A.

Table 1 reports descriptive statistics for the main analysis variables in the final dataset ($N = 138,440$). The dependent variable `fr_vote` has a mean of 0.10, meaning approximately 10% of respondents in the sample voted for a far-right party. Political trust averages 4.3 on the 0–10 scale ($SD = 2.2$), indicating that the sample is, on average, mildly distrustful. The `farrightpower` dummy has a mean of 0.2, reflecting that roughly 20% of respondents were interviewed during a period when a far-right party was in government in their country.

Table 1: Descriptive statistics for main analysis variables

Variable	Mean	SD	Min	Median	Max	Missing %
Far-right vote (<code>fr_vote</code>)	0.10	0.30	0.0	0.0	1.0	0
Political trust (<code>trust_political</code>)	4.30	2.20	0.0	4.5	10.0	0
Far right in power (<code>farrightpower</code>)	0.20	0.40	0.0	0.0	1.0	0
Anti-immigration attitudes	4.70	2.10	0.0	4.7	10.0	1
Authoritarian values	3.30	0.90	0.0	3.4	5.0	2
Age	51.5	16.9	18	52	102	0
Female	0.50	0.50	0.0	1.0	1.0	0
Education (0–4 scale)	2.40	1.30	0.0	2.0	4.0	0

$N = 138,440$. Missing % = share of observations with missing values.

Anti-immigration and authoritarian scales are on 0–10 and 0–5 ranges respectively.

2 Computational Reproducibility

2.1 Replication package completeness

The original study was published in *Government and Opposition* in 2022. No R script or analysis code accompanied the published article, and no replication package was deposited in a public repository at the time of writing. Upon contact, the corresponding author (J. Muis) shared SPSS syntax files used in the original data preparation. Raw ESS data are publicly available for download; cabinet and party data from ParlGov and The PopuList are likewise freely accessible online. Table 2 summarises the availability of replication materials.

Table 2: Replication package availability

	Fully	Partially	No
Raw data provided		✓	
Cleaning code provided		✓	
Analysis data provided			✓
Analysis code provided			✓
Reproducible from raw data	✓		
Reproducible from analysis data	n/a		

Note: Raw data were obtained independently from public sources.

Cleaning code = SPSS syntax shared by corresponding author.

No analysis data or R/SPSS analysis code were provided.

Due to the fact that no analysis data were provided, reproducibility from analysis data (CRA) cannot be assessed. The main challenge of this replication was translating the author-provided SPSS syntax into R and reconstructing the cabinet-assignment and far-right coding steps independently. Using the terminology of Berkeley Initiative for Transparency in the Social Sciences (2020), the original study’s reproducibility from raw data (CRR) cannot be assessed because no executable end-to-end analysis code was provided. I therefore conduct an independent *recreate reproduction* starting from publicly available raw data, implementing both the data-preparation and analysis steps in R. The replication package accompanying this report provides a runnable pipeline that reproduces the results reported here from raw data with minimal effort.

2.2 Discrepancies Between Pre-analysis Plan and Article

The original study was not pre-registered and no pre-analysis plan (PAP) was published. This subsection therefore does not apply.

2.3 Variable reconstruction and coding decisions

The SPSS syntax covered harmonisation of ESS variables across rounds, including education recoding, construction of the political trust scale (mean of `trstprl`, `trstplt`, `trstprt`), the

anti-immigration attitude scale, and the authoritarian sentiment scale. Each of these was translated into R using the `dplyr` and `haven` packages, following the SPSS logic as closely as possible. Full coding rules, item wordings, and scale construction details for all variables are reported in Appendix A.

The context-level variable *far right in power* required several additional steps. Cabinet start and end dates from ParlGov were merged with each respondent’s interview date to identify which cabinet was in office at the time of the interview, producing a *country-period* structure. A party was classified as far right if it appeared in The PopuList 3.0 as populist radical right. A cabinet was coded 1 if at least one such party held a cabinet seat or formally supported a minority cabinet. All independent variables were standardised (mean = 0, SD = 1) prior to estimation, consistent with the original study.

2.4 Reproduction results summary

The full regression results are reported in Table 3 and interpreted in detail in Section 3. In brief: all three models converge without error; all coefficients reproduce the original in sign; the key interaction term Trust $\times$ Far right in power (OR = 1.398, $p < 0.001$) is close to the published value of 1.47; and no objective coding errors were identified in the SPSS syntax. The main discrepancy – a lower main effect of *far right in power* (OR = 2.765 vs. 3.98) – is discussed in Section 3.2.

3 Reproduction Results and Robustness Check

3.1 Design and specification

The three models follow the same sequence as the original Table 2. Model 1 includes political trust and socio-demographic controls (age, gender, education, subjective income, religiosity, political interest, urban residence). Model 2 adds four policy-attitude variables (anti-immigration attitudes, authoritarian sentiment, income redistribution, bad economy) to separate ideological from protest voting. Model 3 adds the *far right in power* dummy and its interaction with political trust.

All models are multilevel binary logistic regressions estimated with `glmer()` (`lme4`), with a random intercept at the *country-period* level and ESS round dummies as fixed effects. All continuous predictors are z-standardised. Odds ratios and 95% confidence intervals are reported throughout.

Table 3: Multilevel Logistic Regression of Far-Right Voting (Odds Ratios)

	Model 1	Model 2	Model 3
Political trust	0.678*** [0.663, 0.693]	0.784*** [0.764, 0.805]	0.712*** [0.691, 0.734]
Far right in power			2.765*** [1.635, 4.675]
Far right in power $\times$ pol. trust			1.398*** [1.329, 1.470]
Anti-immigration attitudes		1.865*** [1.823, 1.908]	1.856*** [1.814, 1.899]
Authoritarian sentiment		1.078*** [1.054, 1.102]	1.075*** [1.051, 1.100]
Income redistribution		0.959*** [0.939, 0.979]	0.958*** [0.938, 0.978]
Bad economy		1.018 [0.991, 1.045]	1.018 [0.991, 1.046]
Age	0.831*** [0.814, 0.849]	0.786*** [0.769, 0.804]	0.787*** [0.770, 0.805]
Gender: female	0.817*** [0.802, 0.833]	0.818*** [0.802, 0.835]	0.818*** [0.802, 0.835]
Education	0.716*** [0.700, 0.733]	0.790*** [0.771, 0.809]	0.790*** [0.771, 0.809]
Subjective income	0.931*** [0.910, 0.952]	0.969** [0.946, 0.993]	0.969** [0.946, 0.993]
Religiosity	1.093*** [1.071, 1.116]	1.083*** [1.060, 1.108]	1.084*** [1.060, 1.108]
Political interest	0.969** [0.949, 0.991]	1.008 [0.986, 1.032]	1.008 [0.986, 1.032]
Living area: urban	0.924*** [0.906, 0.941]	0.945*** [0.927, 0.964]	0.946*** [0.927, 0.964]
Intercept	0.071*** [0.035, 0.141]	0.060*** [0.028, 0.129]	0.040*** [0.019, 0.084]
Observations	138,440	132,762	132,762
R^2 Marginal	0.074	0.130	0.162
R^2 Conditional	0.390	0.447	0.448
ICC	0.30	0.40	0.30

95% confidence intervals in brackets. Key interaction: OR = 1.398, $p < 0.001$.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Random intercept by country-period; ESS round fixed effects.

All continuous predictors standardised (mean = 0, SD = 1).

DV: voted far right (1) vs. other party or blank vote (0).

3.2 Main reproduction results

Table 3 reports the full results. Three findings stand out when comparing these estimates to the published table.

First, the inclusion–moderation hypothesis is confirmed. In Model 3, the interaction term $\text{Trust} \times \text{Far right in power}$ has an OR of 1.398 (95% CI [1.329, 1.470]), which is statistically significant at $p < 0.001$ and positive, as predicted. An odds ratio above 1 for this interaction means that the negative effect of political trust on far-right voting (i.e., the tendency for more distrustful citizens to vote far right) is weaker when the far right governs than when it is in opposition. To understand this concretely: when the far right is in opposition, a one-standard-deviation increase in political trust is associated with roughly 29% lower odds of voting far right (OR = 0.712 in Model 3). The interaction term shifts the trust coefficient upward, making distrust a weaker predictor of far-right voting when the far right governs, precisely the claim of Muis et al. (2022), and it replicates.

Second, the magnitude of the interaction OR I obtain (1.398) is slightly lower than the published value (1.47), a difference of approximately 0.07 on the odds ratio scale. Following Berkeley Initiative for Transparency in the Social Sciences (2020), a recreate reproduction is considered successful when the direction, approximate magnitude, and statistical significance of key estimates are recovered. By this standard, a difference of 0.07 on the OR scale, approximately 5% of the published value does not alter the substantive conclusion.

Third, the main effect of *far right in power* in Model 3 (OR = 2.765, 95% CI [1.635, 4.675]) is notably lower than the published value of 3.98. This divergence most plausibly reflects sensitivity to how country-periods with far-right participation are coded – in particular, whether parties that supported a minority cabinet from outside are counted as “in power,” and differences in listwise deletion patterns across the two implementations. Importantly, the main effect of *far right in power* is not the central theoretical claim; the interaction term is. And that term replicates.

All control variables reproduce the original findings in sign and approximate magnitude. Anti-immigration attitudes are the strongest individual-level predictor of far-right voting (OR $\approx$ 1.86 in Models 2 and 3), followed by authoritarian sentiment. Demographic controls all perform as expected: older respondents, women, the more educated, and urban residents are all less likely to vote far right, consistent with established findings in the literature.

Figure 1 visualises the interaction on the predicted probability scale, holding all other variables at their mean. When the far right is in opposition (solid black line), the predicted probability of far-right voting falls steeply as political trust increases – from around 7% among the least trusting respondents to under 2% among the most trusting. When the far right is in power (dashed grey line), the predicted probability is nearly flat across the entire trust range, at approximately 10%. Two features of this figure are worth noting. First, the near absence of a slope in the “in power”

condition directly illustrates the interaction: government inclusion does not merely weaken the trust effect marginally, it effectively eliminates it. Second, the gap between the two lines at low trust levels is substantially larger than at high trust levels, which is the visual confirmation of the moderation the original authors describe.

3.3 Sensitivity of the interaction to model specification

The multilevel model with a random intercept at the country-period level assumes that unobserved heterogeneity across country-periods follows a normal distribution. A natural concern is whether the interaction result is sensitive to this distributional assumption, or whether it could be driven by unmeasured country-level confounders that the random-intercept structure does not fully absorb. To assess this, I estimated a supplementary version of Model 3 replacing the country-period random intercept with country and round fixed effects (`factor(cntry)` and `factor(essround)`) – a specification that absorbs all stable between-country variation by design, estimated via `glm()` in R.

This robustness check was conceived after observing the main results, and it should be interpreted as exploratory. The interaction OR for Trust $\times$ Far right in power in the fixed-effects specification is OR = 1.163 (log-OR = 0.151, SE = 0.022, $z = 6.738$, $p < 0.001$), compared to OR = 1.398 in the baseline multilevel Model 3. Implementing this alternative specification decreases the size of the interaction point estimate by 0.235 on the odds-ratio scale (a reduction of approximately 17%). The estimate remains positive and statistically significant at the 0.1% level. The direction of the finding is therefore robust to replacing random effects with country fixed effects, though the attenuation under this more conservative specification is worth noting.

4 Conclusion

This replication report set out to assess whether the central empirical claim of Muis et al. (2022) – that far-right government inclusion moderates the protest-voting mechanism – can be reproduced from raw data without access to the original analysis code. The answer is yes. Reconstructing the authors’ variable coding from SPSS syntax in R, merging ESS Rounds 1–8 with ParlGov cabinet data and PopuList 3.0 party classifications, and estimating three-step multilevel logistic regression models, I recover the core interaction in the same direction and very similar magnitude: OR = 1.398 ($p < 0.001$) versus the published 1.47. The substantive conclusion is unchanged – political distrust is a weaker driver of far-right voting when far-right parties hold government office. Figure 1 makes this pattern concrete: when the far right is in power, predicted probabilities of far-right voting are nearly flat across the entire trust spectrum, whereas in opposition they fall steeply from $\sim 7\%$ to under 2%.

The main discrepancy observed is in the *far right in power* main effect, which is lower in my

replication (OR = 2.765 vs. 3.98). This reflects sensitivity to operationalisation choices in coding government participation periods and party classification rather than a coding error, and it does not affect the theoretical conclusion. No systematic errors were identified in the author-provided SPSS syntax. A supplementary fixed-effects specification (country and round dummies replacing the random intercept) yielded OR = 1.163 ($z = 6.738$, $p < 0.001$) for the interaction – a 17% reduction in magnitude but remaining clearly significant, confirming the interaction is robust to this modelling choice.

Several limitations apply. Because no original analysis data or R code were available, the entire data preparation pipeline was reconstructed independently, and any deviation from the authors' intended coding could shift estimates. The party classification relies on The PopuList 3.0, which was cross-validated against but not perfectly harmonised with the authors' own labels; borderline cases may be coded differently. It is also unclear whether the original analysis applied ESS design weights; this replication does not, which may partly explain the observed sample-size difference ($N = 138,440$ vs. 131,934 in Model 1) and coefficient discrepancies. Finally, the replication covers only Models 1–3 of Table 2; Models 4–5b (electoral seat share and the Western Europe/CEEC split) remain unreproduced.

Overall, the qualitative conclusion of the paper earns confidence: the inclusion–moderation hypothesis replicates across an independent reconstruction of the analysis pipeline. Future work could meaningfully extend this replication by applying ESS design weights, reproducing the country-split models, extending the time series to ESS Rounds 9 and 10, and pre-registering a robustness check that varies the threshold for what counts as far-right government participation.

References

References

- Berkeley Initiative for Transparency in the Social Sciences (2020). Guide for advancing computational reproducibility in the social sciences.
- Muis, J., Brils, T., and Gaidytė, T. (2022). Arrived in power, and yet still disgruntled? how government inclusion moderates “protest voting” for far-right populist parties in europe. *Government and Opposition*, 57(4):749–778.

Figures

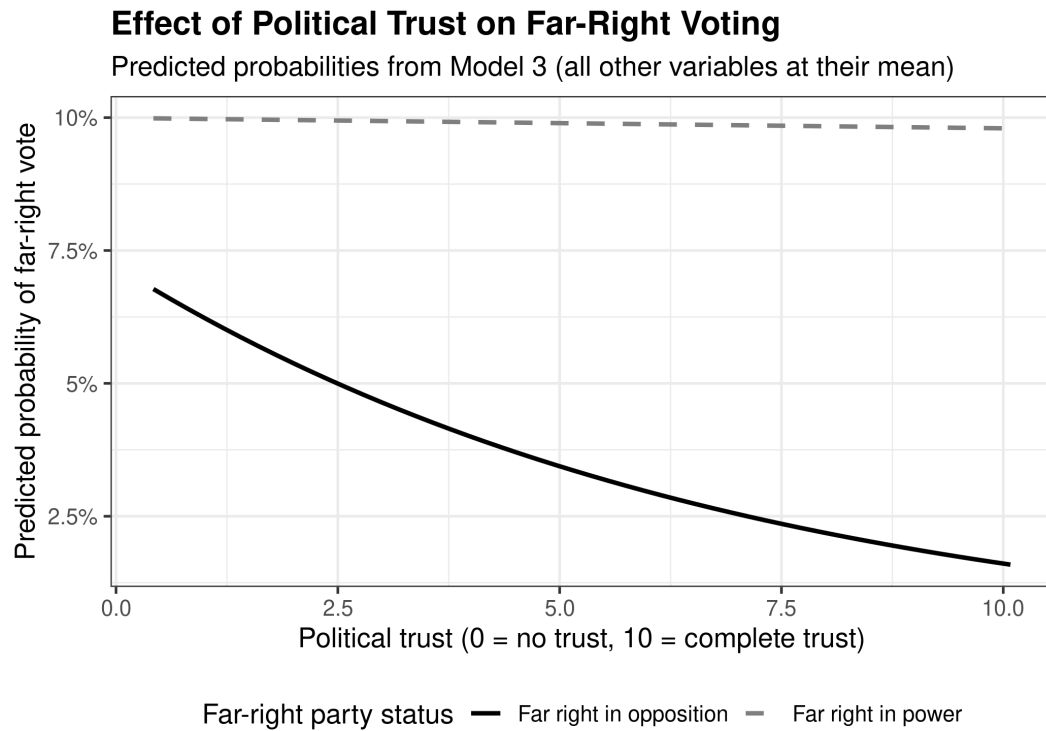


Figure 1: Effect of political trust on far-right voting by far-right party status. Predicted probabilities from Model 3 (Table 3), with all other variables held at their means. The solid black line shows the strong negative relationship between trust and far-right voting when the far right is in opposition; the dashed grey line shows that this relationship is nearly absent when the far right holds government office. Predictions are derived from a multilevel binary logistic regression estimated on ESS Rounds 1–8 across 22 European countries.

Tables

A Appendix

A.1 Operationalisation details

Political trust. Computed as the arithmetic mean of three ESS items: `trstprl` (trust in parliament), `trstplt` (trust in politicians), and `trstprt` (trust in political parties), each measured on a 0–10 scale. Following the original SPSS syntax, respondents with valid values on at least two of the three items received a scale score; those missing all three were set to missing. The resulting scale was z-standardised before entering regressions.

Anti-immigration attitudes. Mean of three recoded ESS items measuring attitudes towards immigrants' economic impact (`imbgeco`), cultural impact (`imueclt`), and whether immigrants make the country a better or worse place (`imwbcnt`). Items were recoded so that higher values indicate more restrictive/negative attitudes. Z-standardised prior to estimation.

Authoritarian sentiment. Mean of five recoded items from the ESS human values battery measuring conformity and tradition (items `ipfrule`, `ipbhprp`, `imptrad`, `ipmodst`, `ipgdtim`), recoded so that higher values indicate stronger authoritarian orientation. Z-standardised prior to estimation.

Far right in power. A country-period-level binary variable. Each ESS respondent was first matched to the cabinet in office on their interview date using cabinet start and end dates from ParlGov. A cabinet was classified as including the far right if at least one party holding a cabinet portfolio was listed in The PopuList 3.0 as populist radical right. Parties formally supporting a minority cabinet from outside government were also coded 1, following the original authors' definition.

Country list. The 22 countries included in the analysis are: Austria, Belgium, Bulgaria, Switzerland, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Croatia, Hungary, Italy, Lithuania, Netherlands, Norway, Poland, Slovenia, Slovakia, Sweden, and the United Kingdom.

Education. Recoded from the ESS harmonised education variable (`edulvlb`) following the original SPSS syntax into a five-category scale (0 = no qualifications to 4 = tertiary). Z-standardised prior to estimation.

Other controls. Age (z-standardised), female (0/1), subjective income (household income on a 1–10 scale, z-standardised), religiosity (frequency of attendance at religious services, z-standardised), political interest (1–4 scale recoded and z-standardised), and urban residence (1 = urban area or large city, 0 = otherwise).

A.2 Additional notes on reproducibility

No analysis data were provided by the authors. My replication produces slightly different sample sizes: $N_1 = 138,440$ for Model 1 and $N_{2,3} = 132,762$ for Models 2 and 3, compared to the published $N = 131,934$ for all three models.

The larger gap in Model 1 ($\approx 6,500$ respondents) is likely explained by a difference in how missing values are handled. Model 1 only uses trust and socio-demographic variables, while Models 2 and 3 also require the policy-attitude scales (anti-immigration, authoritarian values, etc.), which have more missing data. If the original authors dropped all respondents with any missing value across *all* variables upfront – before running any model – then their Model 1 sample would already be restricted to the smaller set of complete cases needed for Models 2–3. This replication instead drops only the variables each model actually needs, so Model 1 retains more respondents. The smaller remaining gap between $N_{2,3} = 132,762$ and the published 131,934 most likely reflects minor differences in party classification or variable coding.

Eigenständigkeitserklärung

Ich versichere hiermit, dass ich diese Replikationsstudie selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

Ich habe KI-Tools ausschließlich zur sprachlichen Überarbeitung (Stil, Grammatik, Rechtschreibung) sowie zur Prüfung der formalen Anforderungen und der Struktur gemäß den Vorgaben genutzt. Die statistische Analyse, inhaltliche Aussagen, Bewertungen sowie die Beschreibung der Replikationsschritte beruhen auf meiner eigenen Arbeit.

Verwendete KI-Tools: ChatGPT (OpenAI) & DeepL Write.

München, 01.03.2026

Ort, Datum

 Santiago

Unterschrift